LifeMenu: A Lifestyle-Medicine-Oriented System for Menu-Grounded Restaurant Ordering

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Abstract

Restaurant ordering is a constrained language-to-action problem: users describe dietary needs in flexible natural language, but the system must recommend only dishes available on the current menu. This challenge is especially critical for lifestyle-medicine-oriented requests. In such scenarios, the system must support everyday dietary preferences without making unsupported nutrition claims or recommending nonexistent dishes. In this paper, we present LIFEMENU, a menu-grounded restaurant ordering system that treats the menu as the boundary for both selection and explanation. LIFEMENU decomposes user requests into hard constraints, health-oriented goals, and soft preferences, then returns ranked menu dishes with short evidence-grounded explanations. Human evaluation shows that the local Qwen model in LIFEMENU improves menu-grounded recommendation quality and provides practical support for restaurant-side dietary preference matching under fixed-menu constraints.

1 Introduction

Restaurant ordering is a routine part of daily life. During ordering, users often have specific dietary needs, such as seeking something light, high-protein, blood-sugar-friendly, or free of certain ingredients. The key problem is that these needs are expressed in flexible natural language, while the system can only recommend dishes available on the current menu. Traditional recommender systems often rely on past user-item interactions or large-scale item matching, which makes them less suitable for handling ordering-time constraints and short natural-language requests. Moreover, traditional recommender systems struggle with the massive fixed item sets typical of restaurant menus. They often result in high inference latency and computational costs, while simultaneously demanding extensive private user information. LIFEMENU addresses this gap by matching users' dietary preferences with dishes available on the current menu, reducing the need for user interaction history or social preference data often used in traditional recommender systems.

The challenge is to match natural-language flexibility with a fixed menu. User requests can be diverse and even conflicting, which increases the difficulty of recommendation. At the same time, daily food choices are an important part of lifestyle-medicine-oriented health behavior (Lippman et al., 2024; Reddy et al., 2019), and recent work on nutrition-related LLMs suggests that real-world dietary support requires careful and clear task boundaries (Li et al., 2025). In this paper, such requests are treated as restaurant-ordering preferences rather than clinical advice: terms such as allergen, high-protein, lower-grease, or blood-sugar-friendly are considered by the system when analyzing user requests and recommending dishes. The LIFEMENU system can provide appropriate recommendations without relying on the user's interaction history or social preferences.

We design LIFEMENU, which is a lifestyle-medicine-oriented domain-specific system for menu-grounded restaurant ordering. It treats a fixed real menu as a closed action space, applies priority rules over constraints and preferences, and generates structured Top-N recommendations with short explanations.

This paper makes two main contributions. First, to our knowledge, LIFEMENU is the first domain-specific system for lifestyle-medicine-oriented natural-language restaurant ordering. Second, we build a local system that achieves competitive performance compared with strong online LLMs while

supporting practical restaurant-side deployment: after local deployment, it does not require repeated calls to external models, can reduce service costs, is less affected by network latency, and better protects user privacy.

2 System Design

This section describes the overall architecture and decision process of LIFEMENU, from interpreting dietary requests and enforcing constraint priorities to producing menu-grounded Top-N recommendations.

2.1 Task Definition

LIFEMENU treats the restaurant menu as a closed action space. At a given ordering session, we let the current restaurant menu be denoted as $M = \{d_1, d_2, \ldots, d_n\}$, where each d_i is a directly orderable dish available on the menu. For a user request x, the system returns a ranked list

$$R_N(x, M) = \{(d^{(1)}, e^{(1)}), \ldots, (d^{(N)}, e^{(N)})\}, \tag{1}$$

where each $d^{(k)} \in M$ and each explanation $e^{(k)}$ must be supported by menu evidence. The system must choose from dishes that are currently on the menu.

2.2 Menu Evidence

Each dish consists of attributes, such as the dish ID, dish name, category, bilingual description, ingredient evidence, potential allergens, pungency degree, health cues, and readable tags. These representations allow the system to match dishes against user constraints and preferences. LIFEMENU does not use external nutrition databases or infer clinical nutrient values.

2.3 Request Decomposition and Ranking

LIFEMENU decomposes each user request into three types:

1. Hard constraints, such as avoiding seafood, pork, dairy, mushrooms, or spicy food, denoted by $C(x)$;
2. Health-oriented ordering goals include non-clinical ordering preferences such as light, high-protein, lower-grease, or low-GI when such cues are available in the menu, denoted by $H(x)$;
3. Soft preferences include staple type, taste, cuisine style, portion preference, or meal context, denoted by $P(x)$.

Formularize the logic as:

$$x \mapsto (C(x), H(x), P(x)), \tag{2}$$

The system recommends based on the following ranking order: First, LIFEMENU excludes dishes that violate hard constraints in the current menu. The system then ranks the remaining dishes by health-oriented goals and soft preferences. These components rank the meals by prioritizing health-oriented goals over soft preferences, while simultaneously ensuring overall relevance to the user's request. Finally, the interface outputs a ranked Top-N dish list with brief explanations. This prevents soft preferences from overriding hard constraints. For example, a seafood dish should not be recommended to a user who asks for no seafood, even if it matches a high-protein goal. Health-oriented goals rank feasible dishes, and soft preferences are only considered when they do not conflict with higher-priority requirements.

2.4 System Overview

LIFEMENU is a menu-grounded ordering system designed for natural-language restaurant ordering. Users can express diverse dietary requests, including restrictions, health-oriented goals, and taste preferences. Meanwhile, the system ensures that all recommendations strictly adhere to the restaurant's current menu. Figure 1 shows the overall workflow of LIFEMENU from user request and menu input to ranked dish recommendations.

From the user's perspective, LIFEMENU analyzes a natural-language request in terms of hard constraints, health-oriented goals, and soft preferences, and then recommends a ranked Top-N list of dishes that can be directly ordered from the current menu. Each recommendation is accompanied by a short explanation grounded in menu evidence, helping the user understand why a dish matches the current request. This design keeps recommendations grounded in the current menu, easy to interpret, and practical for local deployment. Figure 2 shows the deployed LIFEMENU web interface for real menu-grounded ordering. Users can enter natural-language dietary needs, view menu dishes with visible tags, and add suitable items to the cart.

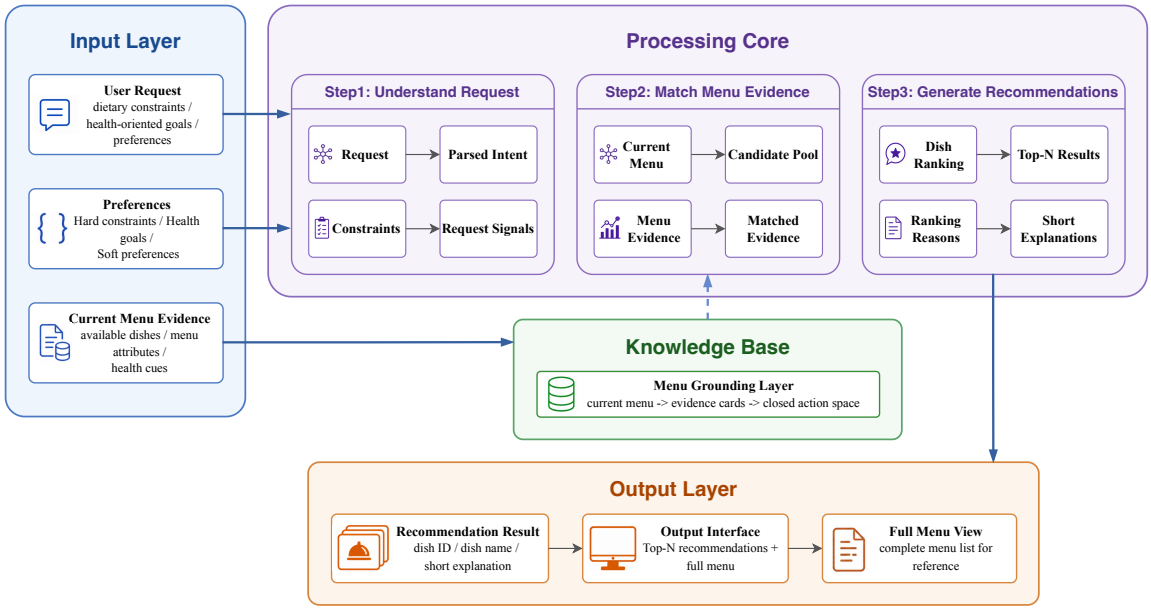


Figure 1: Overview of the LIFEMENU system.

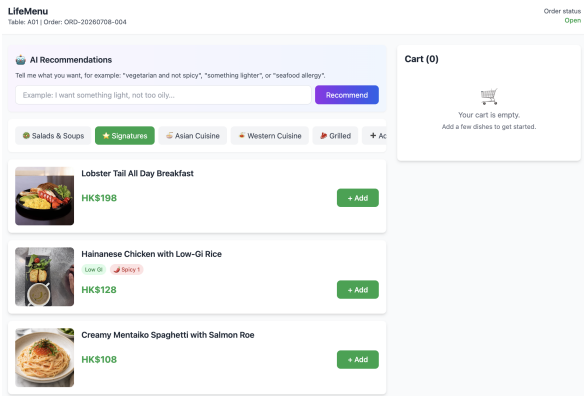


Figure 2: Deployed LIFEMENU demo interface.

3 Data Construction

We construct the LIFEMENU ordering data through three stages, covering natural-language request generation, difficulty-based case classification, and menu-grounded teacher labeling.

3.1 Request Pool Construction

First, we formulated the menu as a closed action space. We then manually designed a taxonomy of restaurant-ordering requests based on common needs and menu-visible dietary cues. This taxonomy covers representative scenarios, preference types, language forms, and meal contexts. These dimensions were polished and completed by LLMs and then used to guide diverse request generation. We then used gpt-5.4-mini to generate a pool of 1,500 candidate requests. After generation, we filtered the request pool to remove duplicates and ensure diversity across multiple dimensions: request expressions, constraint patterns, dish coverage, and explanations. This process yielded a final dataset of 1,000 high-quality requests.

3.2 Standard and Hard Case Classification

Furthermore, we designed a heuristic selector to categorize the requests into two difficulty levels: standard and hard. Standard requests are basic and directed. They do not contain obvious conflicts, and the menu usually has several eligible dishes. In contrast, hard requests include explicit exclusion conditions, conflicting goals, or fewer available dishes. For example, "I want a rice-based meal with steady blood sugar, enough protein, and no seafood or dairy" is a hard request. It is difficult because rice provides carbohydrates, but the user also wants to keep blood sugar steady. The seafood and dairy exclusions further reduce the dishes that the system can recommend. These requests compel the system to make a clearer choice among hard constraints, health goals, and soft preferences. We computed a rule-based difficulty score $s(x)$ for each request to distinguish standard from hard cases, rather than using a separately trained classifier. The score combines four components,

$$s(x) = s_{\text{cons}}(x) + s_{\text{ling}}(x) + s_{\text{scen}}(x) + s_{\text{feas}}(x),$$

where $s_{\text{cons}}(x)$ reflects the number of hard constraints and preference targets, $s_{\text{ling}}(x)$ captures

linguistic markers of conflict, negation, alternatives, and hedging, $s_{\text{scen}}(x)$ reflects the scenario type, and $s_{\text{feas}}(x)$ reflects the number of feasible dishes remaining after hard-constraint filtering. Requests with $s(x) \geq 4$ were labeled as hard, and the others as standard. Applying this procedure to the 1,000-request set yielded 643 standard requests and 357 hard requests.

3.3 Labeling and Data Split

After comparing response quality across several strong online LLMs, we chose gpt-5.4-mini to label answers for all 1,000 requests by returning the Top-N recommended results and explanations for each. Once the teacher model completed the labeling, we evaluated the quality of teacher ordering data through rule checks and manual matching checks. We finally randomly selected 100 of 1,000 entries of request and manually reviewed them based on the following principles: 1. Whether there is any violation of hard constraints. 2. Whether there are appropriate dishes in the list. If all are met, it is a hit, and the final hit rate is 97%.

4 Local Model Training

The goal of local student model training is to transfer the menu-grounded recommendation behavior of the teacher model to a locally deployable 7B model.

4.1 Local Student Adaptation

We use Qwen2.5-7B-Instruct (Qwen et al., 2025) as the base model and adapt it with QLoRA (Dettmers et al., 2023). This parameter-efficient training method allows the model to learn the recommendation pattern within the fixed menu with lower memory cost. Compared with repeatedly calling online LLMs, the local student is more suitable for restaurant-side deployment. Because inference is handled locally, the system is less affected by network latency and does not need repeated calls to external models, which can reduce service costs. It also decreases reliance on external LLMs, thereby offering stronger privacy protection. The target training and deployment environment is a single RTX 5090 32GB GPU.

The training data come from the 800 training samples in the 1,000 teacher-labeled samples described above. Through this training method, the student model learns to return recommendations within a closed action space that meet user constraints, match health goals and soft preferences, and provide explanations grounded in the current menu.

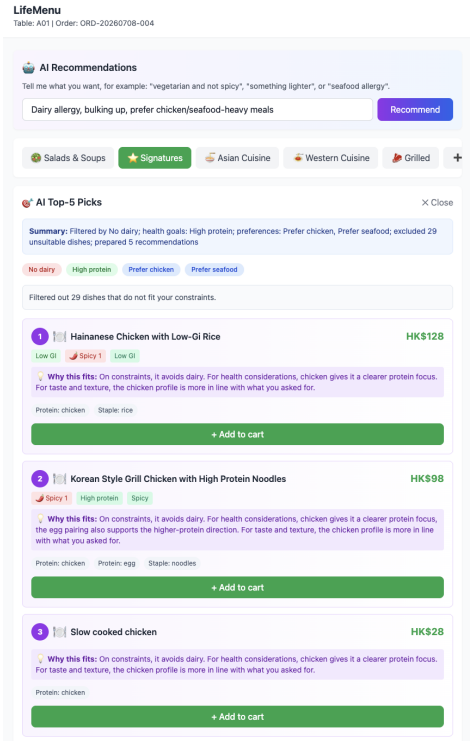


Figure 3: A LIFEMENU recommendation example for a constrained dietary request. The interface presents a grounded Top-5 ranking with request-aware explanations.

During the inference stage, LIFEMENU receives the user's current request and the complete set of menu candidates. The local Qwen student parses the user's requirements and returns the Top-N recommended results, along with brief menu evidence explanations. This reasoning setup enables the system to handle real-world ordering scenarios. Furthermore, it facilitates local restaurant-side deployment: the system is less affected by network latency, avoids repeated calls to external models, reduces service costs, and offers stronger privacy protection.

Figure 3 shows an example of the deployed inference interface. Given a constrained request such as dairy allergy and a preference for high-protein chicken or seafood-heavy meals, LIFEMENU summarizes the parsed needs, filters unsuitable dishes, and returns a ranked Top-N list. Each recommendation is shown with menu-visible tags and a short explanation that links the dish to the user's constraints, health-oriented goals, and taste preferences.

5 Evaluation

We evaluated LIFEMENU under the same fixed-menu setting as student inference.

5.1 Evaluation Setup

The evaluation mainly addressed two questions: First, whether the adaptation of QLoRA could significantly improve the local Qwen2.5-7B-Instruct base model; second, whether the adapted local model could match the recommendation quality of strong online LLMs under the same menu constraints. We input 100 held-out test requests into Qwen QLoRA Student, Qwen Base, Claude Sonnet 4.5, Gemini 2.5 Pro, GPT-4o, and DeepSeek-V3. All systems received the same user requests, menu evidence, and output requirements. Restaurant ordering is a preference-based ranking task, where several dishes may be reasonable for the same request and the best order can depend on personal taste and dietary priorities. Therefore, we evaluate perceived recommendation quality through blind human assessment.

5.2 Human Evaluation Protocol

The human evaluation was conducted anonymously. For each request, outputs from the six systems were shown to human reviewers under anonymous system names. The reviewers did not know which model each output came from. The evaluation included 100 requests. For each request, reviewers compared the outputs of the six systems, and each output was scored by six reviewers. We employed six undergraduate students as reviewers.

Reviewers assigned an overall score ranging from 1 to 5 based on five comprehensive criteria. Specifically, they evaluated whether the recommendations (1) matched user needs, (2) complied with hard constraints, (3) adhered to lifestyle-medicine principles, (4) provided a reasonable ranking, and (5) offered menu-grounded explanations. We use an overall score rather than separate scores for each dimension because restaurant ordering is closer to the real user experience: users usually judge the recommendation list as a whole rather than scoring for each component. Therefore, this score better captures the practicality of the final Top-N list in the ordering scenario.

After scoring, we unblinded the model identities and computed three metrics for each model. *Overall* represents the average score across all 100 test requests, *Standard* is the average on the standard request subset, and *Hard* is the average on the hard request subset.

5.3 Main Results

Table 1 summarizes the main results. In this table, Overall is the average human rating across all test requests, while Standard and Hard report scores on the two request subsets. The 95% CI shows the uncertainty range of each Overall score. Student – Model is the paired score difference between Qwen QLoRA Student and each baseline on the same requests; Paired 95% CI and p indicate the uncertainty and statistical significance of this difference. Higher scores are better.

Qwen QLoRA Student achieves the highest Overall average score of 4.444, with a 95% confidence interval of [4.318, 4.570], indicating that the adapted local model can generate high-quality recommendations in the fixed-menu ordering setting.

The improvement over Qwen Base is substantial. Qwen Base obtains an Overall score of 3.108, while Qwen QLoRA Student reaches 4.444. The paired difference is +1.336, with a paired 95% confidence interval of [1.132, 1.539] and $p < 0.00001$. This statistically significant gap shows that QLoRA adaptation improves the base model's ability to handle menu-based recommendation requests.

Compared with strong online LLMs, the student model remains competitive. Its Overall score is close to Claude Sonnet 4.5 and Gemini 2.5 Pro, and modestly higher than GPT-4o and DeepSeek-V3, by about 3.9% and 5.8%, respectively. This competitive performance suggests that adaptation on menu-based labeled data helps the student model better understand restaurant-ordering requests and produce constraint-aware recommendations.

The Standard and Hard columns further show performance across request types. Claude Sonnet 4.5 obtains the highest score on Standard requests, while Qwen QLoRA Student achieves the best score on Hard requests, with an average of 4.464. This suggests that task-specific adaptation is especially helpful for complex ordering requests with stronger constraints, multiple dietary preferences, or competing needs.

Overall, Qwen QLoRA Student shows promising results in the menu-grounded restaurant-ordering setting. It substantially improves over the unadapted Qwen Base model and remains competitive with strong online LLMs, especially on hard requests. These results indicate that a locally adapted student model can learn useful constraint-

aware recommendation behavior from menu-based labeled data. In Appendix B., two case studies further demonstrate how the student model handles ingredient-level constraints, health-oriented goals, and user preferences in concrete ordering scenarios.

6 Related Work

LIFEMENU is related to nutrition-oriented language systems, LLM-based recommendation, efficient adaptation, and grounded interfaces.

6.1 Food Recommendation and Dietary Support

Recommender systems match users with relevant items by modeling preferences, context, and constraints (Adomavicius and Tuzhilin, 2005). In the food domain, prior work has surveyed healthy-food recommenders (Trang Tran et al., 2018), broader food recommendation frameworks (Min et al., 2020), recipe healthiness for meal planning (Trattner and Elsweiler, 2017), and constrained QA over structured food knowledge (Chen et al., 2021). Together, these studies show that food recommendation must consider ingredients, dietary goals, restrictions, and user context, not only taste.

6.2 Lifestyle-Medicine and Nutrition-Oriented Language Systems

Lifestyle medicine emphasizes nutrition and other lifestyle factors in long-term health management (Lippman et al., 2024), and prior medical discussions call for stronger integration of nutrition and lifestyle medicine into practice and training (Reddy et al., 2019). This motivates our focus on everyday dietary goals such as lighter meals, higher-protein options, and blood-sugar-friendly staples. Recent work has evaluated LLMs in nutrition-related language tasks and dietary support settings (Li et al., 2025). Unlike generic nutrition-oriented generation, LIFEMENU grounds such requests in dishes available on the current restaurant menu.

6.3 LLM-Based Recommenders and Grounded Interfaces

LLMs have been studied for recommendation (Ren and Huang, 2025), recommender agents (Peng et al., 2025), conversational preference optimization (Tajiri and Inaba, 2025), explainable recommendation (Ma et al., 2024), and user-centered explanation refinement (Zhang et al., 2025). These studies are relevant because restaurant ordering requires request understanding and interpretable explanations.

Prior work also maps natural-language requests into structured representations for task-oriented systems. SUQL translates conversational questions into formal queries over structured and unstructured data (Liu et al., 2024), while schema-based state tracking and SQLWOZ represent dialogue states, constraints, and preferences (Xu et al., 2025b,a). LIFEMENU follows this direction by organizing ordering requests into constraints, health-oriented goals, and preferences, then selecting dishes from the current menu.

7 Conclusion

We presented LIFEMENU, a lifestyle-medicine-oriented system for menu-grounded restaurant ordering. The results show that a locally adapted student model can handle constraints, preferences, and menu-grounded explanations and remain competitive with strong online LLMs in a fixed-menu setting. Future work will test LIFEMENU across more diverse restaurant menus, where cuisines, ingredients, allergens, and health-related cues may vary substantially.

Limitations

LIFEMENU is designed for ordering-time dietary preference matching, not for clinical nutrition advice. The current system mainly relies on menu text, menu-visible evidence, user request, and menu-derived cues, so its recommendations should not be interpreted as medical-grade dietary guidance. It does not estimate precise nutrients, diagnose health conditions, or recommend treatment. In addition, our current evaluation is based on one restaurant menu, while different types of restaurants may provide very different dishes, ingredients, and cuisine styles.

Ethics and Broader Impact Statement

LIFEMENU is intended to support restaurant-ordering preferences, not to provide medical or clinical nutrition advice. A key ethical risk is that incorrect recommendations may cause harm if they violate hard constraints, such as allergens, vegetarian or vegan requirements, halal or other religious dietary restrictions, or explicit ingredient exclusions. This risk is especially important when menu information is incomplete, outdated, or inconsistent with actual kitchen preparation. To reduce

Model	Overall	95% CI	Standard	Hard	Student–Baseline Diff.	Paired 95% CI	p
Qwen QLoRA Student	**4.444**	**[4.318, 4.570]**	4.429	**4.464**	–	–	–
Claude Sonnet 4.5	4.422	[4.302, 4.540]	**4.443**	4.396	+0.022	[-0.134, 0.177]	0.781
Gemini 2.5 Pro	4.355	[4.219, 4.489]	4.351	4.360	+0.089	[-0.090, 0.268]	0.328
GPT-4o	4.278	[4.139, 4.415]	4.319	4.227	+0.166	[-0.003, 0.335]	0.054
DeepSeek-V3	4.200	[4.038, 4.359]	4.214	4.182	+0.244	[0.046, 0.443]	0.016
Qwen Base	3.108	[2.929, 3.288]	3.086	3.136	+1.336	[1.132, 1.539]	< 0.00001

Table 1: Blind human evaluation on 100 held-out test requests. Overall scores are averaged over six human raters. Student – Model denotes the paired difference between Qwen QLoRA Student and each baseline. Higher scores are better.

this risk, LIFEMENU treats hard constraints as the highest-priority signals, grounds recommendations in menu-visible evidence, and avoids unsupported nutrition claims. In real deployment, restaurants should maintain accurate ingredient and allergen information, and users with severe allergies or strict dietary requirements should verify the final order with restaurant staff. Because dietary preferences may reveal sensitive health, religious, or lifestyle information, local deployment can also reduce unnecessary exposure of user requests to external model services.

Human raters were informed of the evaluation purpose, participated voluntarily, and evaluated only system outputs without collecting personally identifiable user information.

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Appendix A. Prompt Templates

This section summarizes the main prompt templates used for request generation, teacher labeling, and test-time evaluation. Dynamic fields are replaced with placeholders for readability. We report the key instructions rather than the full runtime prompts to keep the appendix compact, while preserving the main constraints used in each stage.

A. Request-generation prompt. The request-generation prompt asks `gpt-5.4-mini` to generate bilingual, natural user requests for LIFEMENU. The prompt requires each request to sound like a real user typing into a menu recommendation system, rather than a label-like query. It also asks the model to avoid dish names, direct recommendations, repetitive openings, translation-like Chinese, and overly abstract wording. Dynamic diversity fields are used to improve coverage across health scenarios, preference types, language forms, meal contexts, and request lengths.

```
Request-generation key instructions:
- Generate bilingual user requests only.
- Do not generate dish names or recommendations.
- Requests should be natural, concrete, and grounded in
  state, goal, preference, or restriction.
- Use realistic restaurant-ordering language, such as
  not too oily, less sauce, not fried, with rice,
  with vegetables, soup, grilled, or not too spicy.
- Keep English and Chinese aligned in meaning, but avoid
  word-for-word translation.
- Avoid repetitive phrasing and label-like queries.
- Cover diverse health scenarios, preference types,
  language forms, meal contexts, and request lengths.
- Return strict JSON.
```

The generated requests are not used as final answers. They provide diverse user-side inputs for later teacher labeling. This separation keeps request generation and recommendation labeling as two different steps.

B. Teacher recommendation prompt. The teacher prompt asks `gpt-5.4-mini` to produce Top-5 labels from filtered menu candidates at temperature 0. The goal is to obtain menu-grounded supervision for training the local student model. Since hard-constraint-violating dishes are already removed before ranking, the teacher focuses on ordering the remaining candidates according to the user request, health-oriented goals, explicit preferences, and general suitability.

```
You are a menu-grounded teacher recommender for LifeMenu.
Select Top-5 dishes from the provided menu candidates only.

Rules:
- Hard-constraint-violating dishes have already been removed.

- Never mention excluded dishes.
- Rank candidates by the user request, health/lifestyle
      goals,
  explicit preferences, and general suitability.
- Use only the provided menu candidates.
- Recommend exactly five unique dishes when possible.
- Explanations must be concise, positive, and grounded in
      menu evidence.
- Do not invent ingredients, nutrition facts, calories,
  medical claims, or cooking styles.
- Avoid negative fallback phrasing such as not a noodle dish,
       but...
- Output JSON with top5 and parsed_intent fields.

Input fields:
User request: <user_request>
Hard constraints removed from menu: <hard_constraints>
Reference health goals: <health_goals>
Reference preference targets: <preference_targets>
Menu candidates:
D01 | ...
D02 | ...
...
```

The teacher output contains two parts: a ranked recommendation list and a parsed-intent summary. The recommendation list provides dish IDs, dish names, ranks, and short explanations, while the parsed-intent fields record hard constraints, health-oriented goals, soft preferences, and ambiguities.

C. Shared test-time prompt. During evaluation, all compared models receive the same user request, menu candidates, and output requirements. This shared prompt ensures that the comparison focuses on recommendation quality rather than differences in input format. The local student model is also tested with the same format for fair comparison.

```
You are LifeMenu, a menu-grounded restaurant recommendation
      assistant.
Given a user request and a fixed menu, recommend exactly
      five dishes.

Input:
User request: <user_request>

Menu candidates:
D01 | <dish_name> | category: <category> |
ingredients: <ingredient_evidence> |
tags: <readable_tags> |
health_cues: <explicit_health_cues>
...

Rules:
- Recommend exactly five dishes.
- Use only dish IDs from the given menu.
- Preserve hard constraints.
- Do not invent dishes, ingredients, calories,
  nutrition numbers, or medical claims.
- Return JSON with top5 and parsed_intent fields.
- Explanations must be grounded in menu evidence.
```

This shared test-time format is used only to standardize evaluation inputs and outputs. It does not give any model access to hidden labels or human scores.

Appendix B. Case Studies

Standard case: ingredient-level exclusion. Table 2 shows a standard case where the user asks for no garlic and prefers a grilled option. This case tests whether the system can preserve an ingredient-level constraint while still satisfying a preparation

preference. To keep the comparison compact, the table reports dish outputs only, while explanations are discussed in the text.

Model	Score	Top-5 dishes
Qwen QLoRA Student	**5.0**	**1.** D16 Argentine Beef Rib Eye; **2.** D17 Argentine Beef Sirloin; **3.** D19 US Angus Short Rib; **4.** D20 Seared Barramundi with green oil; **5.** D01 Seared Salmon & Mushroom with Curry Sauce and Low-GI Rice.
Qwen2.5-7B Base	3.0	**1.** D15 Pork With Green Peppercorn Sauce; **2.** D16 Argentine Beef Rib Eye; **3.** D17 Argentine Beef Sirloin; **4.** D18 Lamb Chop with Garlic & Rosemary; **5.** D30 Hainanese Chicken with Low-GI Rice.
GPT-4o	4.0	**1.** D16 Argentine Beef Rib Eye; **2.** D17 Argentine Beef Sirloin; **3.** D19 US Angus Short Rib; **4.** D20 Seared Barramundi with green oil; **5.** D32 Salmon and Jap Sakura Ebi Rice with Soup.
Claude Sonnet 4.5	4.0	**1.** D15 Pork With Green Peppercorn Sauce; **2.** D16 Argentine Beef Rib Eye; **3.** D17 Argentine Beef Sirloin; **4.** D19 US Angus Short Rib; **5.** D20 Seared Barramundi with green oil.

Table 2: Case study on REQ_0164 (standard): *"No garlic today, and I'd like something grilled if there's a choice."* The table reports dish outputs only.

Qwen QLoRA Student receives the highest score because its ranked list stays focused on grilled or seared main dishes and avoids clear garlic conflicts. Its explanations emphasize the preparation style and suitability of the selected dishes, such as describing beef steak options as grilled and straightforward, and describing the seared fish option as a lighter grilled-style choice. In contrast, Qwen2.5-7B Base includes D18 Lamb Chop with Garlic & Rosemary, which directly conflicts with the no-garlic request. Claude also ranks D15 first, even though its explanation notes that garlic appears in the menu evidence for the accompanying vegetables. This case shows that menu-grounded evidence is important: a dish may appear suitable from its name or category, but still violate the request when ingredient details are considered.

Hard case: preparation constraint and fish preference. Table 3 shows a hard case combining a preparation constraint, *no fried food*, with a fish preference and a grilled-style preference. This case is harder because the model must first preserve the hard constraint and then keep the ranking focused on the preferred food type. The request also includes a conditional preference, because grilled fish is preferred if such an option is available.

Model	Score	Top-5 dishes
Qwen QLoRA Student	**5.0**	**1.** D20 Seared Barramundi with green oil; **2.** D01 Seared Salmon & Mushroom with Curry Sauce and Low-GI Rice; **3.** D25 Caesar Salad with Smoked Salmon; **4.** D32 Salmon and Jap Sakura Ebi Rice with Soup; **5.** D21 Smoked Salmon.
Qwen2.5-7B Base	2.5	**1.** D20 Seared Barramundi with green oil; **2.** D01 Seared Salmon & Mushroom with Curry Sauce and Low-GI Rice; **3.** D24 Kale and Tomato with Pomegranate; **4.** D27 Creamy Mushroom Soup; **5.** D30 Hainanese Chicken with Low-GI Rice.
GPT-4o	4.0	**1.** D01 Seared Salmon & Mushroom with Curry Sauce and Low-GI Rice; **2.** D20 Seared Barramundi with green oil; **3.** D32 Salmon and Jap Sakura Ebi Rice with Soup; **4.** D25 Caesar Salad with Smoked Salmon; **5.** D21 Smoked Salmon.
Gemini 2.5 Pro	3.5	**1.** D20 Seared Barramundi with green oil; **2.** D01 Seared Salmon & Mushroom with Curry Sauce and Low-GI Rice; **3.** D32 Salmon and Jap Sakura Ebi Rice with Soup; **4.** D33 Korean Style Grill Chicken with High Protein Noodles; **5.** D18 Lamb Chop with Garlic & Rosemary.

Table 3: Case study on REQ_0226 (hard): *"No fried stuff tonight, and I'd rather have fish if there's a grilled option."* The table reports dish outputs only.

Qwen QLoRA Student keeps all five recommendations centered on fish or seafood options that do not use an explicitly fried preparation. Its explanations also preserve the request priorities: the top recommendation is described as fitting the no-fried preference, the salmon-and-rice option is framed as a balanced choice, and the final smoked salmon item is explained as a simple, light, fish-forward option. By contrast, Qwen2.5-7B Base starts with two suitable fish dishes but then drifts to vegetables, soup, and chicken rice, weakening the fish preference across the ranked list. Gemini 2.5 Pro also relaxes the fish preference by introducing grilled chicken and lamb in the fourth and fifth positions. Together, the two cases show that the adapted student model improves concrete recommendation behaviors, including constraint preservation, menu-evidence checking, and priority-aware ranking.